

Baseline subtraction of wide-band spectroscopic data using least-squares fit by cubic Hermite spline with translational-symmetric basis

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ABSTRACT:

Increasing capability of receivers and spectrometers allows us to obtain a large amount of spectroscopic data related to millimeter-/submillimeter-astronomy with high frequency resolutions and wide frequency ranges. Since it is unlikely that an obtained data set of wide-band spectrum is free from baseline warps and/or ripples, useful methods of baseline subtraction are desired. We propose a new baseline subtraction method using least-squares fit by a uniform cubic Hermite spline. The number of nodes is the only parameter to be specified. This method is similar to cubic spline fit, though the second-order derivative of cubic Hermite spline is not continuous at the nodes. The model function is represented as a linear combination of symmetric and antisymmetric localized basis functions, and this basis has translational symmetry. The fitting parameters are function values and derivatives at the nodes. Thus, basis functions and the fitting parameters each have a clear meaning. This method can be applied in an automatic manner to a whole spectrum with signals of spectral lines masked. We demonstrate the method by analyzing a data set of spectra obtained with the Spectrometer Using superconductor Mixer Receiver (SUMIRE).

Motivation

- Subtracting baseline ripples of wideband spectra
- Models for baseline
 - polynomials
 - trigonometric functions
 - piecewise polynomials

→ Cubic spline (cspline) fit is a conventional method [1],
but we propose another method here.

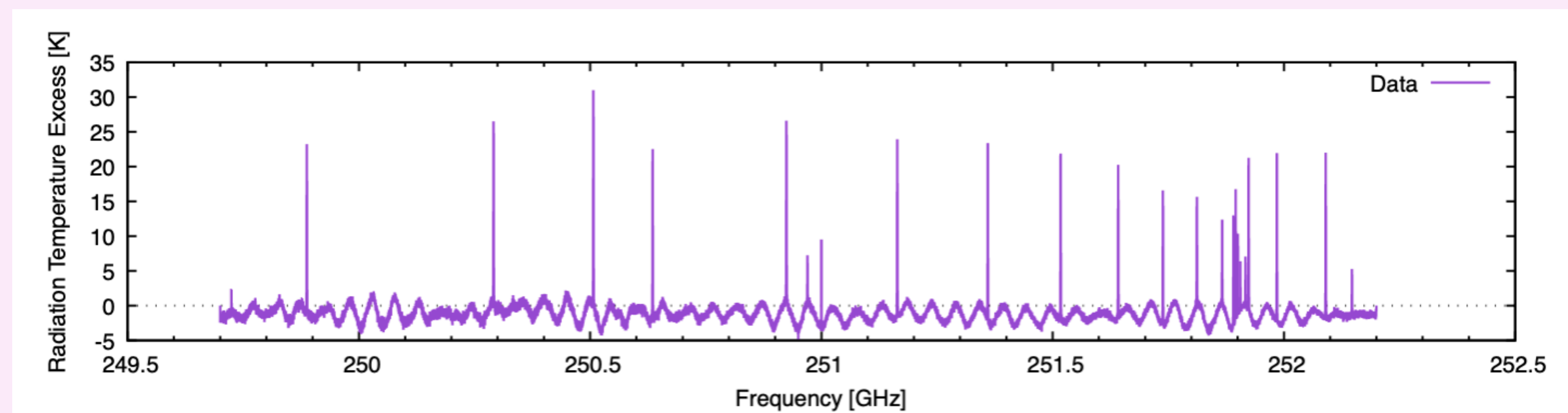
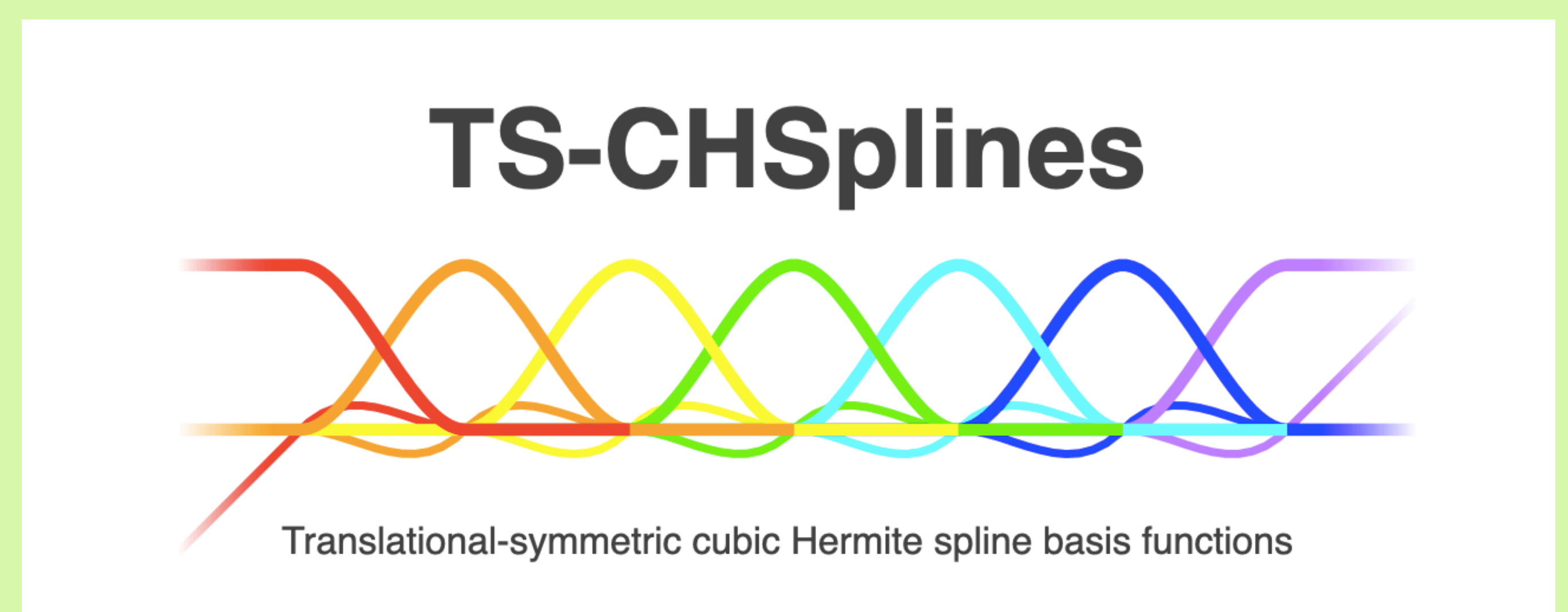
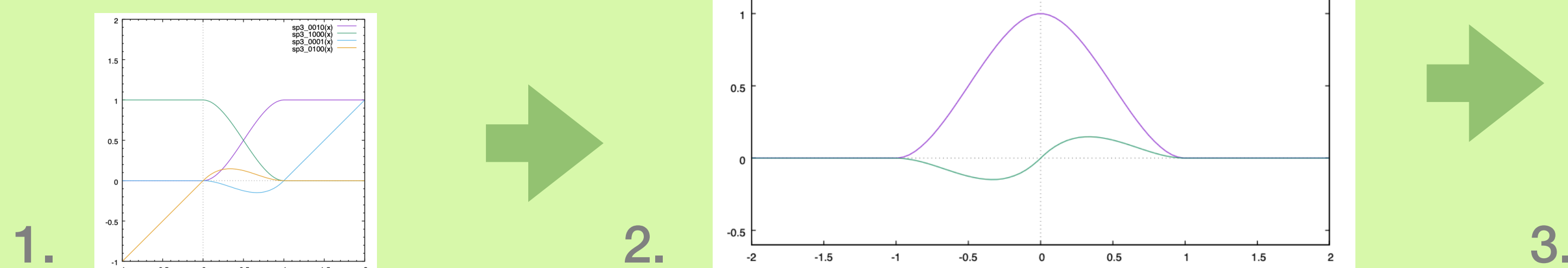


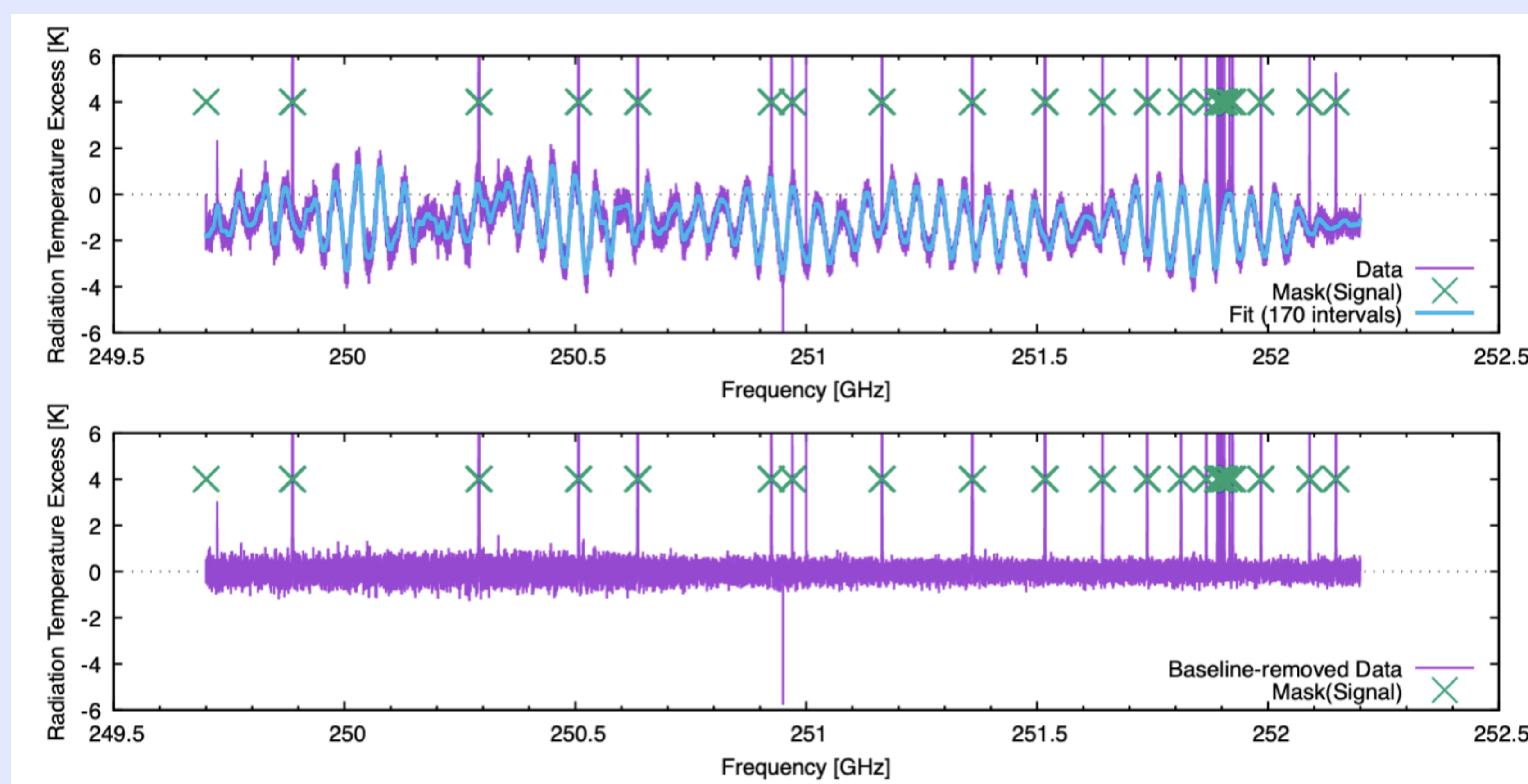
Fig. 1: Example of a wideband spectrum obtained with SUMIRE [2] (before an upgrade of lens with anti-reflection)

Concept

1. Cubic Hermite spline (CHSpline): cubic splines with the Hermite basis [3],
2. Symmetric and antisymmetric basis functions
3. Basis with translational symmetry



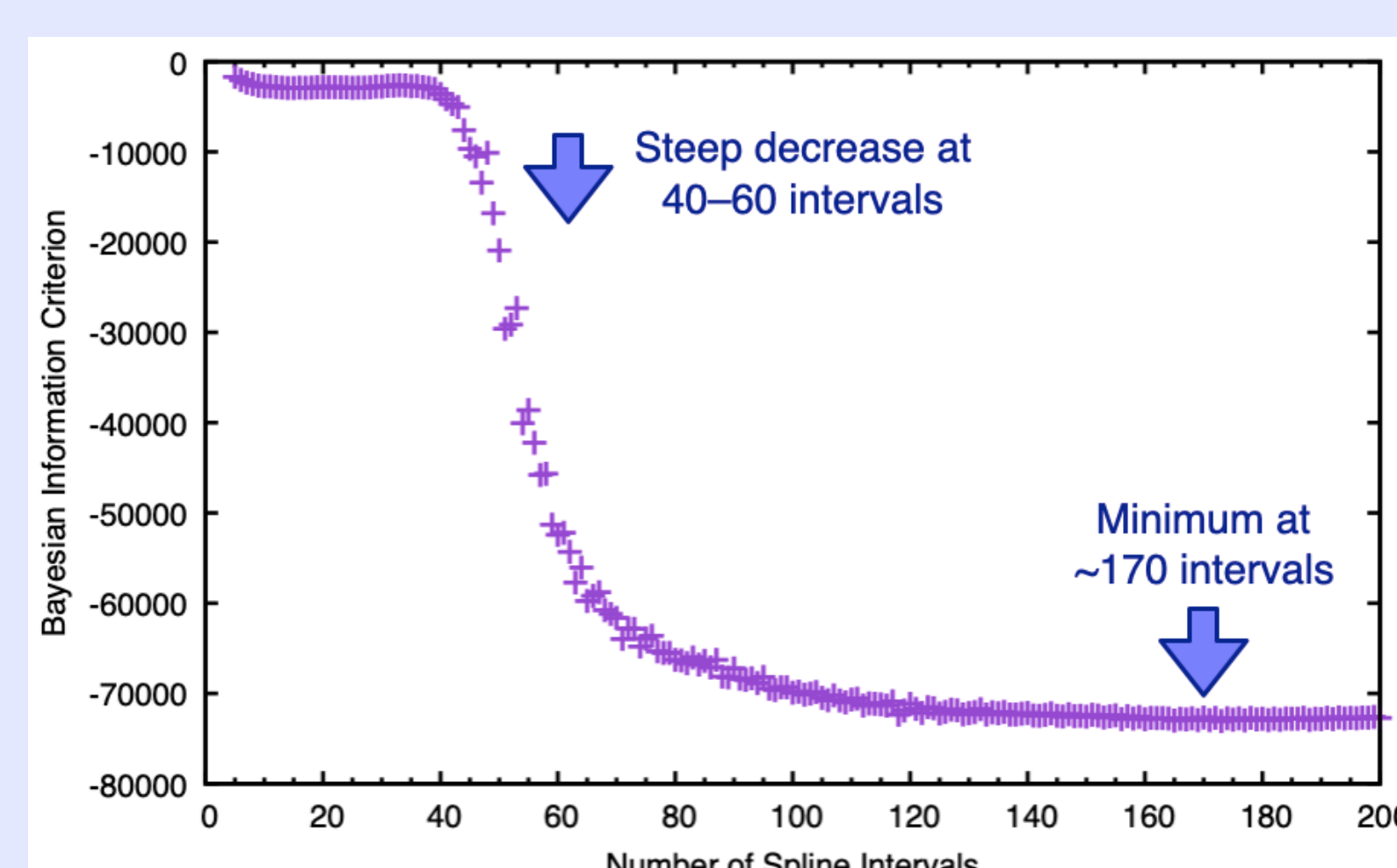
Example



Fit result for Fig. 1 data. The fit model has 170 intervals (171 nodes).

The baseline ripple is traced by the model
function of TS-CHSplines!

Plot of Bayesian Information Criterion (BIC) [4]
to decide the number of spline intervals for
Fig. 1 data.



Implementaion

- A python implementation can be found in a
GitHub repository.

<https://github.com/Nagai-Makoto/ts-chspline>



Future work:

- Make quantitative comparison with
conventional cubic spline fit.

References:

1. CASA User Reference & Cookbook (2013)
2. Watanabe+. PASJ **73**, 2 (2021)
3. Amat+, J. Sci. Comput. **95**:84 (2023)
4. Stoica & Selen, IEEE Signal Process. Mag. **21**, 4 (2004)